

Assignment – 16:

Problem Statement:

To accept a list of N integers and partition the list into two sublists containing even and odd numbers.

Code:

```
#include <stdio.h>

int main()
{
    int n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d integers:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Even numbers:\n");
    for(i = 0; i < n; i++)
    {
        if(arr[i] % 2 == 0)
        {
            printf("%d ", arr[i]);
        }
    }
    printf("\nOdd numbers:\n");
    for(i = 0; i < n; i++)
    {
        if(arr[i] % 2 != 0)
```

```
    {  
        printf("%d ", arr[i]);  
    }  
}  
}
```

Input & Output:

```
Enter number of elements: 10  
Enter 10 integers:  
34 56 79 99 12 13 51 76 52 90  
Even numbers:  
34 56 12 76 52 90  
Odd numbers:  
79 99 13 51
```

```
Enter number of elements: 5  
Enter 5 integers:  
2 3 5 1 132  
Even numbers:  
2 132  
Odd numbers:  
3 5 1
```

